Bosnia and Herzegovina

Academic year 2024-2025.

International University of Sarajevo

PROJECT REPORT

Electrical Distribution in a Restaurant: Real-Life Principles and Scaled Prototype Implementation

Course: Electrical circuits I

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Sarajevo, June 2025. **Ibrahim Čustović 230302273**

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1. Introduction

Electricity in every system in which people live has long been no longer a luxury but a necessity. One of the obvious examples of this is a restaurant where every segment of those businesses is dependent on electricity, from people choosing where to stay and spend their money based on the environment they are in, which is regulated by ambient lighting, then the proper operation of kitchen appliances that ensures that the food reaches the table in the right condition, not spoiled or cold, and then that they stay in an environment that is safe for them and that implements all the safety measures that are provided for this type of space, which is impossible to achieve without proper installed electronics. All the previously mentioned items tell us that the restaurant is one big system that could not work without electricity, and we will talk more about how it works below.

My task for this project was to create a functional restaurant model, with electrical systems that include lighting, ventilation, and some type of sensor, without using any advanced technology such as microcontrollers or the like, just control with switches. After creating the model, I had to write a report that included the process of making it, as well as presenting the model itself through explanations and pictures, and how it looks in reality, starting from the power source to some of the smaller details that ensure the proper operation of the entire system and provide a good foundation for its proper functioning and maintenance.

2. Electrical Wiring in a Restaurant Model

2.1 Idea

The initial idea was to make a system of 3 rows of LEDs, 2 motors as a cooling and ventilation system that will be turned on by separate switches, a small breadboard that represented a socket for devices in the kitchen as well as a flame detector and all together to be connected to a power bank from a phone that provides a voltage of 5V. Since my soldering is very bad, and the system is a bit too complicated to connect to one big breadboard, I decided to fix everything with hot glue, however, it turned out to be not a very smart idea, since some of the components lost contact easily and even the slightest mistakes, removing the component from the hot glue can be an impossible job. For lighting, we used 3 rows of LEDs, red for the outside of the restaurant (7 pieces), 4 pieces of white for the kitchen and 8 pieces of green and yellow in combination for the restaurant hall. Since I was thinking of connecting everything to one power source, I thought that the previously mentioned power bank enters all the switches via the main switch that would serve or imitate the operation of the fuse. The only component that should always be on is the flame sensor, which should work as long as it receives voltage. However, most power banks have protection against undervoltage, sudden oscillations, or an unrecognized source, and therefore I could not start small DC motors via the power bank because they draw a lot more current than they use when starting, so they make sudden oscillations. There are some tricks like connecting another LED in parallel with the motors, but I did not succeed, so I connected the motors to a separate power source of two AA batteries that give a total of 3V, which was the perfect solution for this project. In the rest of this section I will explain the components I used, how I connected them, the problems I encountered, and the things I would have done differently.

2.2 Components Used

Figure 1: Switches

2.2.1 Switches

Mini switches are small mechanical components that allow or interrupt the flow of electricity when pressed. They work like simple ON/OFF switches. They are most often used to control input signals in electronic circuits. They have two or more pins that connect to an electrical circuit. In this case, I used the mini switches shown in the picture with 2 pins, or Single Pole Single Throw (SPST). When closed, they allow the flow of electricity through the connected wires.

Figure 2:Copper Insulated Wire

2.2.2 Copper Insulated Wire

This wire has a copper core that conducts electricity well. Around the copper is a plastic insulation that prevents unwanted contacts and short circuits. It is used to connect components in electronic and electrical circuits. It can be single-core or multi-core, depending on the need. It is flexible and suitable for working on prototype boards and projects where the "wire is in the air" like this one. I bought mine at a local flea market, so I am not sure what the exact thickness of the wire is, although I think it is somewhere around 1mm.

Figure 3: 3V DC Motors

2.2.3 3V DC Motors

These are small electric motors that operate on a direct current of about 3 volts. When connected to a power source, they begin to rotate due to electromagnetic interaction. The speed of rotation depends on the voltage and the load. They are used in small projects such as toys, fans or mini robots. They need sufficient current and are often connected via transistors or motor drivers. In this project, I used motors directly connected to the voltage source, using one diode connected in parallel with them (or in reverse parallel, but since the direction of rotation of the motor was not important to me, it doesn't matter which term we use).

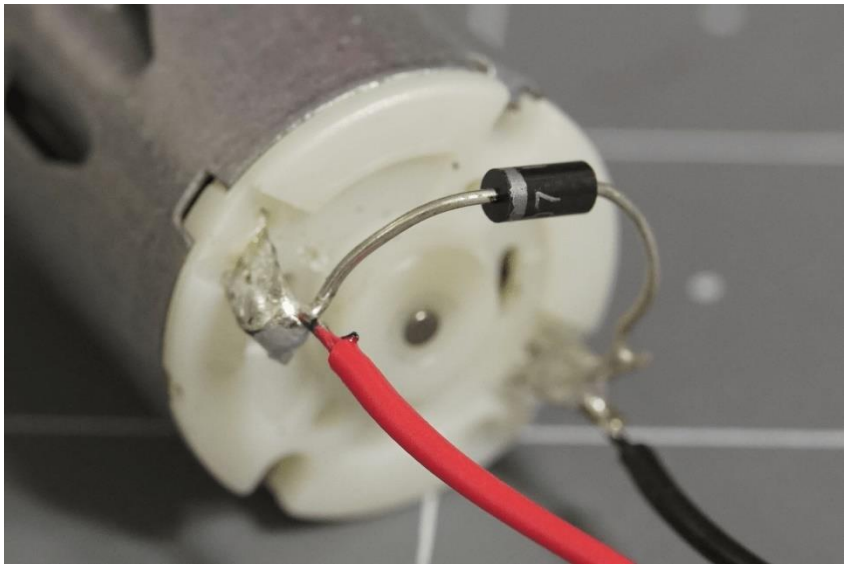


Figure 4: Flyback Diode on Motor

2.2.4 Flyback Diode

Diodes that protect motors are called flyback diodes or protective diodes. They are placed in parallel with the motor, but in reverse polarity, so that they normally do not conduct current. When the motor is turned off, due to its inertia, it produces a reverse (inductive) voltage, which can damage transistors or the microcontroller. The flyback diode then redirects this reverse current and protects the rest of the circuit. Without it, an inductive surge can cause sparks, damage, or malfunctions in the circuit. Even if it is just a simple electrical connection between the motor and the switch, as is the case with us, the flyback diode still has an important protective function. When the motor is running, a magnetic field is created in it. When we break the connection (turn off the switch), this field quickly collapses and the motor generates a short but strong reverse voltage. This voltage can cause a spark on the switch and damage it over time. The flyback diode, placed in reverse parallel with the motor, takes this surge and safely dissipates it, preventing sparking and wear on the switch.

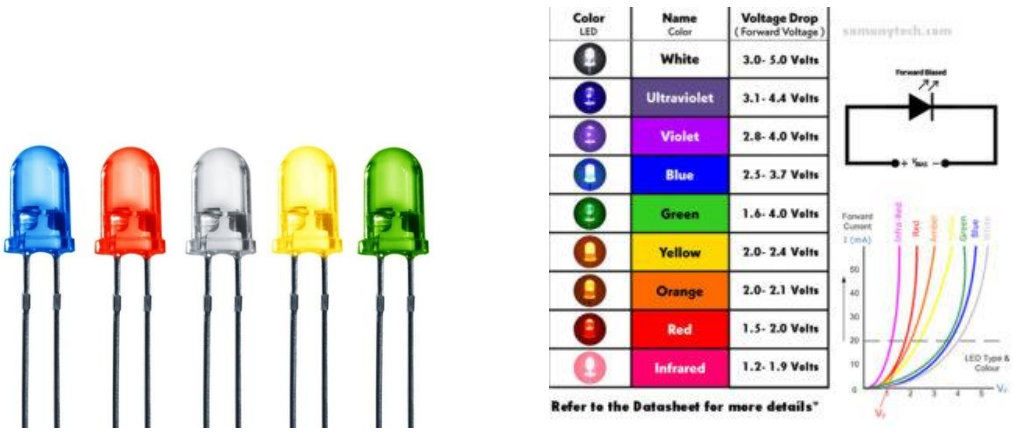


Figure 5: LEDs

2.2.5 LED Diodes

LED (Light Emitting Diode) is a semiconductor diode that emits light when current passes through it. Current can only flow in one direction – from anode to cathode. If connected the other way around, the LED does not light up because it does not conduct. It usually needs about 2V to work,

and is used with a resistor to limit the current. It comes in various colors and is used for signaling and lighting. Until this project, I did not know that the voltage drop of each color of diode is different, so in order for them to light up the same, we need to use different resistor values if we want them to shine with approximately the same intensity. For this project, this was not necessary, so I used a 220Ω resistor for each one and thus additionally protected each LED, ensuring that they last longer.

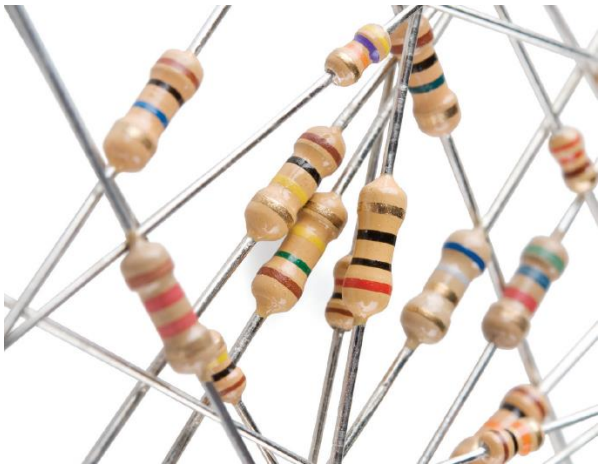


Figure 6: Resistors

2.2.6 Resistors

Resistors limit the flow of electrical current in a circuit. They have a defined resistance value expressed in ohms (Ω). They protect other components, such as LEDs, from excessive current. They do not emit light or move mechanically - they are passive components. They are placed in series in a circuit to reduce voltage or current. As we will see below and as we have previously noted, each of the LEDs in this circuit is connected to a 220Ω resistor.

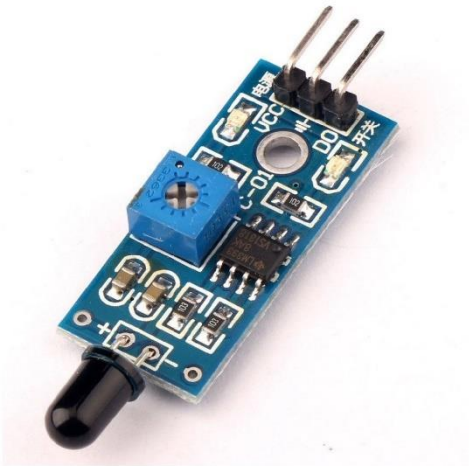


Figure 7: Flame Sensor

2.2.7 Flame Sensor

A flame sensor is an electronic component that detects the presence of a flame or the light emitted by a fire. It is most often used in security systems to automatically shut down or warn when a flame is present or disappears. There are several types: photodiode, phototransistor, UV and IR sensors, and all respond to specific wavelengths of light coming from a fire. In typical Arduino or microcontroller

projects, a flame sensor provides a digital (0/1) or analog signal, indicating whether a flame is present or not. It is placed at a safe distance from the fire source and can respond to a lighter, candle or burner. For this project, I used an IR sensor that sends 0V if activated so I connected it to the minus pole of the buzzer and the positive pole of the buzzer directly to the power source.

Figure 8: Buzzer Types

2.2.8 Buzzer

A buzzer is an electronic component that produces sound when an electric current passes through it. It can be active (it has a built-in oscillator and plays when you apply voltage to it - used in this project) or passive (you have to give it a signal of the appropriate frequency yourself). It is most often used for alarms, notifications, signals and sound effects in projects. It operates on low voltage (usually 3–5V) and can be controlled by a switch, transistor or microcontroller. It is connected so that one pin goes to the power supply, and the other through a resistor or transistor to ground.

Figure 9: Jumper Wires

2.2.9 Jumper Wires

Jumper wires are used for quick and easy connection of components on a breadboard. They have metal pins that are easily inserted into holes in the board. There are male and female connectors, depending on the application. They do not require soldering, so they are ideal for experimentation. They come in different colors for easier tracking of connections. I used them to connect a flame sensor because of their protruding pins, the female side of the jumper wire was ideal for the flame

sensor as well as the pins on the buzzer. I also used them to connect a socket simulation and a mini fridge that I plugged into that socket (mini breadboard).

Slika 10: Mini Breadboard

2.2.10 Mini Breadboard

A mini breadboard is a small plastic board for temporarily connecting electronic components without soldering. Beneath the surface are metal connectors that connect the holes in the rows, allowing for easy circuit assembly. It is ideal for small prototypes, simple tests, and education. Power supplies and components can be easily inserted and removed, allowing for quick error correction. Although it is smaller than a standard breadboard, it functions the same - it usually has 17 rows of 5 holes, without side (voltage) strips. In this case, we brought the voltage and ground to one row in parallel, so we had 4 mini sockets if we connected them to the breadboard with jumpers.

3. Model Development

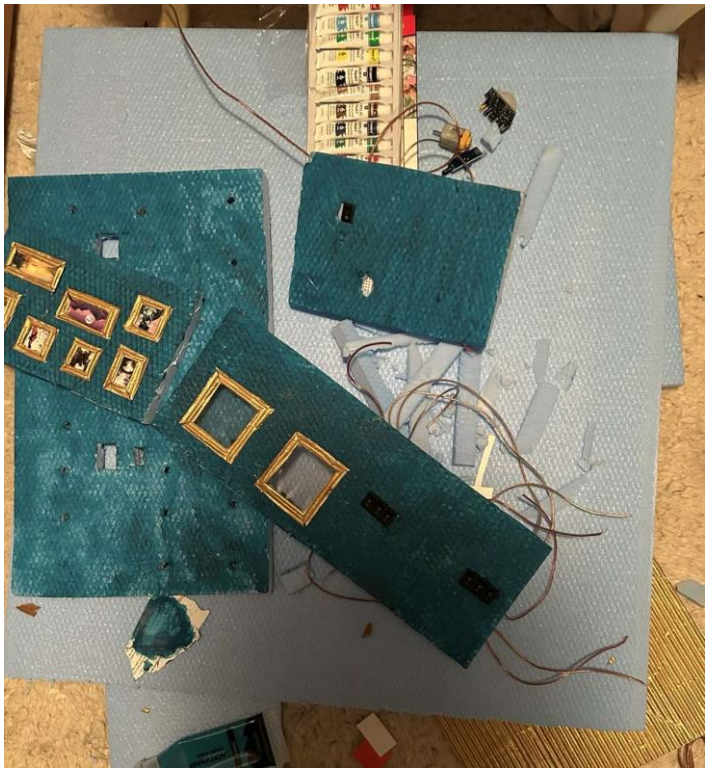


Figure 11: Model Structure Making

3.1 First Step: Choosing Materials and Developing Model Structure

The first step was deciding what materials to use and the actual procurement of materials and tools I needed, which in this case was a hot glue gun. For the structure itself, or the shell of the restaurant building, I decided to use XPS, which is similar to EPS but crumbles less and clean cuts are a little easier to make and is a little stronger than EPS itself. In addition, I had to decide what dimensions the building would have, so I started with the base of the model, which I decided to make 50x30, and based on the fact that I planned to have two layers of styrofoam, I roughly estimated how much space I would need for it, so I started measuring and cutting the material. For painting the walls and floor, I used tempera and acrylic paints, and for decorating the window frames and picture frames, I used the inside of the cardboard, the ribbed part that we discover when we peel off the first layer, and painted it with gold spray paint. After I finished this part, I determined where the switches would be and cut the holes for them, and glued everything except the roof into one piece.

Figure 12: Preparing LEDs

3.2 Second Step: Connecting Motors and LEDs into Subcircuits

After making the base of the model, I moved on to the electrics, and I connected all the switches with pieces of wire and glued them to the model. In addition, there was also the problem of preparing the LEDs for the model. Since their legs were slightly shorter than 2 cm (about 1.8 and 1.6 cm), I had to add some wires, so I added a little copper wire to the negative side, and I extended the positive side by adding a 220 ohm resistor that definitely had to be there in the connection to protect the LEDs, so this was an additional function for them, you can see this in the picture above.

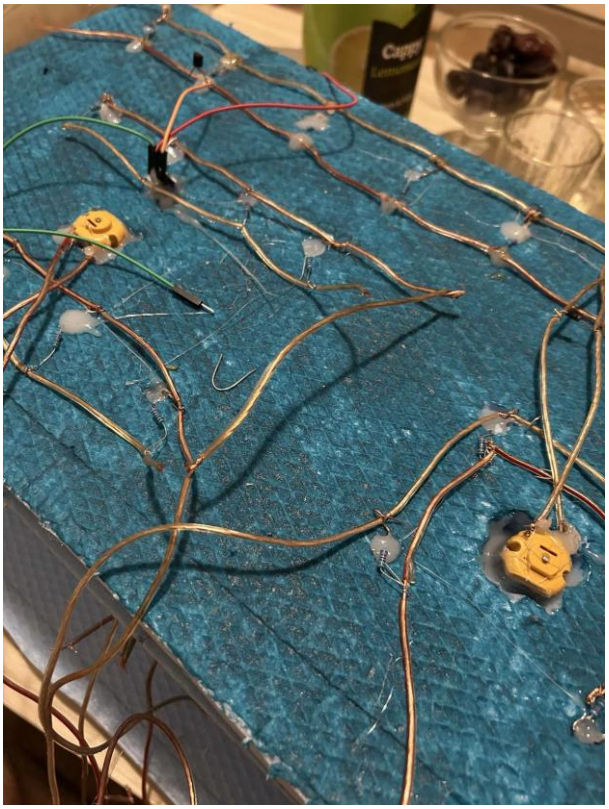


Figure 13: Connecting Subcircuits

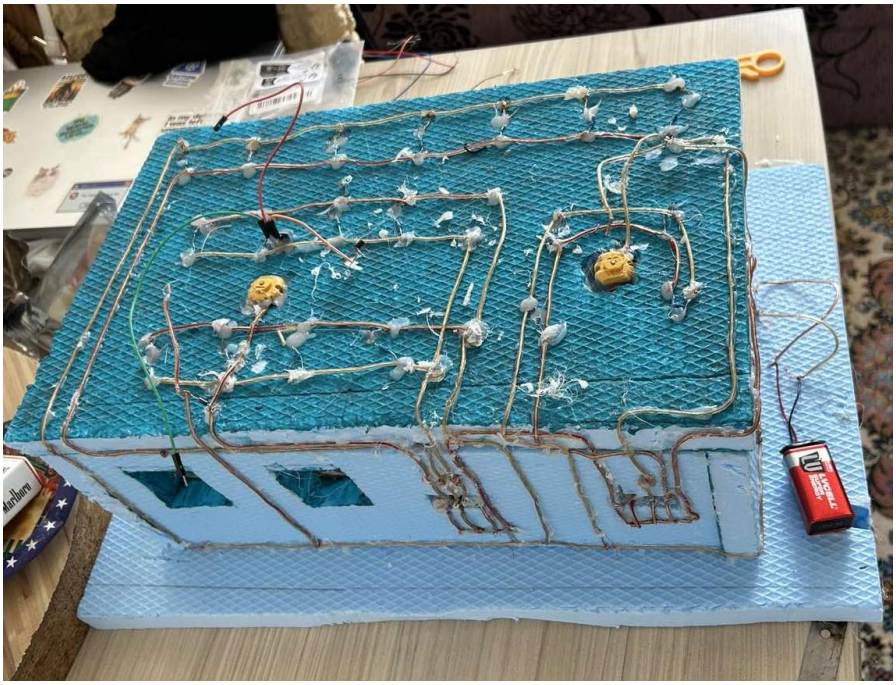
3.3 Third step: Connecting Subcircuits to Main Power Source Through Switches

After ensuring that the branches of the leds are long enough and protected by resistors, I had to connect all parts of the circuit in a parallel connection, one with the other, as well as the connections with each other. I achieved this by having each element directly access both plus and minus. In addition, I also connected one protective diode in reverse parallel to the motors, in order to protect it from sudden drops and rises in voltage when starting and stopping the motor itself.

Slika 14: Connecting Everything Into One Whole

3.4 Fourth step: Connecting Remaining Components

After connecting all the LEDs and motors, I moved on to connecting the mini socket and the flame sensor via jumper wires, of which I used only one side, and they had to be connected in parallel. For the output signal from the flame sensor, I used a buzzer whose positive side I connected directly to the battery voltage, and the minus to the sensor output, because in case of receiving stimulation from the environment, the flame sensor sends 0V, which closes the circuit and activates the buzzer.



Slika 15: Final Wiring Look

After I developed all the connections, it was time to connect the main source to the main switch and further to the other switches, and therefore all the parts of the circuit after the voltage source as well as the ground. It looked very nice at the beginning, while I planned to connect everything

to one voltage source (in the picture above), however, as I said before, the motors cannot be started via a power bank, so I had to separate them into two sources, which in the end looked a lot messier, because of the two voltage vods as well as the ground.

Slika 16: Final Enterior Look

3.5 Fifth step: Last Touch

The last step was to add some smaller elements such as a refrigerator that I just made out of cardboard to try out the operation of the outlet simulation by connecting the resistor and the LED to the jumper cables and plugging it into the cardboard (pictured below on the left). Unfortunately, I don't have a picture of the fridge turned on because I'm not in Sarajevo right now because of the holidays. Also, I added a thin styrofoam armor used to decorate surfaces, mostly ceilings, which only served as a cover to hide the wiring, so it wasn't fixed, it could be removed at any time. On the sides I also added some kind of green paper that has a canvas texture, to imitate grass. I wanted to add furniture, but since I already lost a lot of time on this project, and I also had projects from other subjects, and knowing that it has no crucial importance for the project from this subject, I decided not to do it. In the picture below you can see how the project looked with all the lights on. Unfortunately, I cannot show the operation of the fan in the kitchen and restaurant hall in the picture.

Figure 17: Final Exterior Look

3.6 Problems I Encountered and Things I Would Do Different

Some of the problems I faced were a lack of technical skills, probably because he was dealing with some things for the first time. There were minor injuries on my hands, which I solved by washing the wounds with iodine, so no major problems arose. Also, apart from the problems themselves, I noticed some things that I could have used other methods, for example, right at the start I took out the soldering, although it was not a good idea because gluing with hot glue cannot replace soldering, it can only be an additional method of hardening, in this way I lost contact in some moments and in some parts connection, which should not happen. Also when I lost contact in the motor I had to get it out of the hot glue, which was almost an impossible mission, and when I finally got it out I broke a pin, so I had to change it. Apart from the technical difficulties themselves, time management played a very big role here as well, and apart from the fact that the project could have been of better quality with better team management, the components could have come out much cheaper if I had ordered them from AliExpress or Temu instead of buying from resellers from OLX. I can say that this model was a very good start in this area and that a lesson was learned and that the next project like this could be much easier to plan and implement.

Slika 18: Wounds I Made Making This Model

4. Real-World Restaurant Electrical Systems

4.1 Introduction to Restaurant Electrical Systems

Electrical systems are the basis of the functioning of all modern facilities, and their importance is particularly pronounced in catering facilities such as restaurants. Restaurants are spaces in which various activities take place simultaneously that depend on a stable and safe supply of electricity - from the operation of high-power kitchen appliances, through lighting systems, to air conditioning, ventilation and fire detection systems. Electrical installations in such spaces must be designed and executed according to high standards in order to ensure smooth operation, long-term system life and maximum safety for staff and guests. Given that restaurants operate daily, often more than 12 hours a day, any failure in the electrical network can lead to serious problems - downtime, loss of materials and even a danger to human safety.

In this context, understanding electrical circuits - their construction, division, dimensioning and protection methods - becomes crucial for the proper planning and maintenance of electrical installations in restaurants. When designing a restaurant electrical system, it is necessary to pay attention to the different functional units: the kitchen area, the main guest area, the pantry, the garden (if any), the toilet, the technical rooms and the staff area. Each of these zones has specific requirements in terms of the type of consumers, the required voltage, current and protection. For example, kitchen appliances require high currents and are often connected to three-phase voltage, while lighting consumes less energy and can be low-voltage in decorative zones.

Also, an important part of the electrical system are elements such as switches, fuses, cables, sockets, lamps and control devices. Their proper use and arrangement not only ensure functionality, but also increase efficiency and reduce energy consumption. The introduction of LED lighting, sensor switches and smart control systems is becoming increasingly common in modern restaurants. In addition, adequate protection against fire, overload and short circuits is also necessary, which is achieved by using differential switches and proper grounding.

I think that building this model helped me to research and study this topic in a basic way, which is necessary for simulating this system, but that the real learning began with writing this report, which I believe is an excellent introduction to this topic, explained in a way that even a beginner like me can understand it without any problems.

4.2 Power supply and main distribution

The power supply of a restaurant is the basis of the entire electrical power system of the facility. In most cases, restaurants are connected to the public electricity distribution network via a three-phase connection (3 × 400 V), since a large amount of professional equipment uses three-phase motors and high-power devices. The main distribution of electricity is carried out via a central distribution cabinet (distribution cabinet), which is equipped with automatic switches, fuses and differential protection for all consumer groups in the facility. Inside the main cabinet, the current is further divided into several columns or zones - specifically for the kitchen, guest room or hall, ventilation, lighting, cooling devices, sockets and backup power sources if any (e.g. UPS or generator).

Figure 19: Example of Main Distibution Board

The design of the main distribution must take into account the total power of all consumers and allow for an even distribution of the load across the phases in order to avoid imbalance and overloading of the network. In this regard, the current calculation (calculation of the total current across the phases) is used to size the main cables, switches and fuses. The thickness of the main cable running from the connection to the distribution board depends on the maximum consumption and the distance - most often used cables with a cross-section of 10 mm², 16 mm² or more, made of copper and covered with PVC insulation. In the distribution board itself, there must be a clear marking plan for all electrical circuits, as well as access for quick disconnection in the event of a fire or technical problem.

Figure 20: UPS System

In addition to the main power supply, some restaurants (especially those with larger capacities or those in remote areas) use auxiliary power sources such as diesel generators or UPS systems. Generators allow the continuation of operation of key systems (e.g. cold rooms, lighting, cash registers) during a power outage. UPS systems serve as a short-term backup, most often for computers, sensors, alarms and parts of the automation system. All of these units must be properly integrated into the main distribution board, with automatic or manual power transfer.

In conclusion, the power supply and the main distribution board form the "backbone" of the restaurant's electrical system. Their proper design and sizing enable safe, reliable and energy-efficient operation of all devices and systems that are necessary for the daily functioning of the facility. The next step in the system analysis is a review of the lighting installations, which have both a functional and aesthetic role in the restaurant space.

4.3 Lighting

Lighting in restaurants has a multiple role: it enables visibility and safe work of staff, provides a pleasant atmosphere for guests, and often serves as a decorative element. In the technical sense, lighting can be divided into working (functional), ambient and safety lighting. Work lighting is used in kitchens, warehouses and technical rooms, where strong and focused light is needed. Ambient lighting is used in guest rooms and gardens, with the aim of creating a pleasant atmosphere, while emergency lighting ensures minimum visibility in the event of a power outage (e.g. exit lights for evacuation).

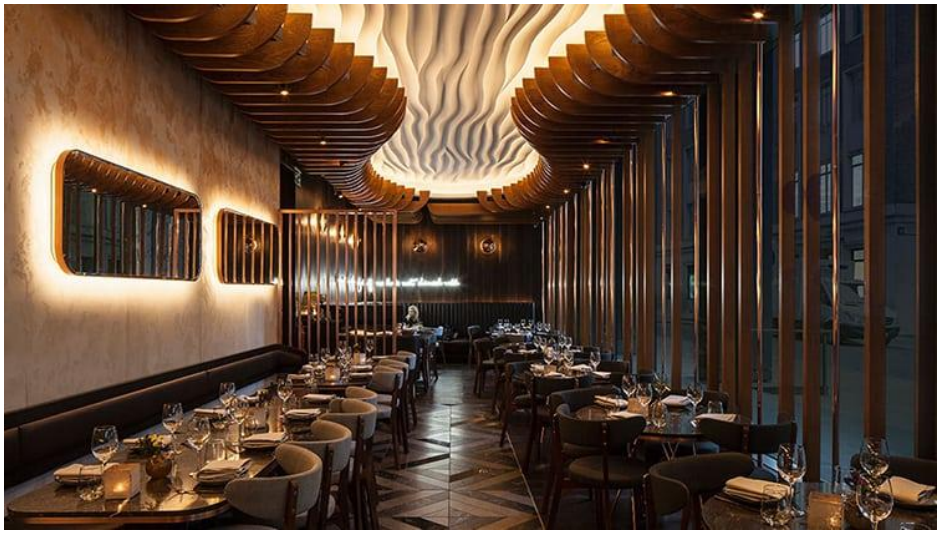


Figure 21: Ambient Lighting of a Restaurant

The most common light source in modern restaurants are LED lamps, which are used due to their low consumption, long service life, low heat emission and the ability to control the light intensity (dimming). In addition to LED lamps, halogen or fluorescent tubes are occasionally used, but less and less due to higher consumption and shorter service life. Lighting is designed to meet lighting standards – for example, a kitchen should have a brightness of 300-500 lux, while a guest room can have a significantly lower value, in keeping with the surroundings. When I was building the restaurant model, I didn't know about this, but somehow it made sense to me, so I used white LEDs to light the kitchen, a combination of green and yellow LEDs to light the guest room, and red LEDs for the front, which were a kind of lamp that emits heat and is used as a kind of heating for the outside of the restaurant.

Figure 22: Restaurant Kitchen LIghting

Lighting control can be simple, using standard switches, or automated - using motion sensors, timers, remote controls and smart systems. Smart lighting systems are increasingly implemented in more luxurious buildings, which allow changing the color, intensity and mode of operation of lamps through a centralized system or application. Such systems can be programmed to change the lighting during the day (eg brighter during lunch, dimmed in the evening), which additionally saves energy and contributes to the aesthetics of the space.

As for the electrical circuits, the lighting is usually carried out using 1.5 mm^2 wires, protected by 6 to 10 A circuit breakers. The lighting circuits are clearly separated from the sockets and motor circuits in the switchboard for better control and safety. Special attention is paid to protection against overload and short circuit, as well as grounding of metal bodies of lighting fixtures. Properly designed and maintained lighting contributes not only to the aesthetics, but also to the functionality and safety of the facility.

Through the analysis of lighting, we can see how theoretical knowledge about electric circuits is directly applied in practice. In the next chapter, we will analyze cables and wire lines, their characteristics and proper use in different areas of the restaurant.

4.4 Electrical sockets and devices

Sockets and electrical devices are one of the most important segments of a restaurant’s electrical system because they directly power the devices needed for daily operation. Restaurants use different types of sockets, depending on the purpose and the required power. Standard Schuko sockets are most commonly used (for devices up to 16 A), but in professional kitchens and technical rooms we also find industrial three-phase sockets (32 A and more), intended for connecting equipment such as electric ovens, steam convection ovens and large refrigerators. Each socket must be properly wired, with appropriate protection and marking in the distribution panel.

Slika 23: Industiral Socket

Sockets are designed to be located in places where there is a real need to connect devices, but also at a safe distance from sources of water, heat or flammable materials. In the kitchen, for example, sockets must be placed in a way that prevents them from being splashed with oil or water. In the guest area, sockets are often hidden (behind furniture) and are used for decorative lighting, ambient music or chargers. All electrical devices connected to the network must be in good condition, with proper grounding and regular protection checks.

A large number of high-powered devices are used in the kitchen - from deep fryers and mixers to refrigerators, freezers and air conditioners. These devices require special electrical circuits with cables with a cross-section of 2.5 mm² to 6 mm², depending on the load. The connection is made through direct connections (without a socket) or through industrial connection systems. It is important that each consumer has its own protective switch in the distribution board, which enables independent disconnection in case of maintenance or failure.

In addition, restaurants use various electronic devices: POS systems, cash registers, music systems, security cameras, ventilation and computers. These devices have lower power requirements, but require a stable and reliable power supply, often with additional surge protection. For them, a special group of sockets with additional protection (eg surge protectors or UPS systems) is usually used.

Electrical outlets and appliances are the user's "points of contact" with the electrical system. Their proper installation, sizing, and protection are essential for user safety and long-term device life. The rest of the report will analyze the wiring structure and the criteria for selecting the thickness and type of cables that connect all elements of a restaurant's electrical system.

4.5 Size And Types of Wires (Wiring)

Choosing the right size and type of wire in restaurant electrical installations is critical to the safe and efficient operation of the electrical system. Every electrical circuit - whether it supplies lighting, sockets, motors or electrical panels - must use conductors of suitable cross-section, material and insulation. Copper wires with PVC insulation are most often used, due to their good conductivity, flexibility and resistance to heat and moisture. The minimum standard wire size for lighting circuits is 1.5 mm², while 2.5 mm² wires are used for sockets. For consumers of higher power (such as ovens or freezers), wires with a section of 4 mm² to 10 mm² are used, depending on the length of the lines and the total load.

The calculation of the wire size is based on the maximum current that will flow through the circuit, the allowable voltage drop and the conditions in which the wire is used (temperature, position - in the wall, in pipes, in the air, etc.). In addition to the cross-section, it is also important that the wires have the correct color according to the standards: phase conductors are usually red, black or brown; zero is blue, and the protective conductor (ground) is yellow-green. This color must be strictly followed to avoid confusion and ensure easy service and maintenance of the system.

Figure 24: PVC Wiring Duct

Installations in restaurants are usually carried out inside plastic pipes (ducts) in walls and floors or surface ducts are used in technical rooms and kitchens. In particularly stressed areas, such as places with high temperature, moisture or the possibility of mechanical damage, cables with additional insulation and mechanical protection are used (eg rubber cables, cables with metal protection or cable tapes with reinforcement). Electrical cabinets must be connected with wires that are resistant to heat and sparks or grease, especially in kitchen areas.

Neglecting the correct choice of cross-section can lead to overheating of the wires, which is a serious fire hazard. Wires that are too thin for the load cause energy losses, voltage drops and a shorter service life of the equipment. On the other hand, wire cross-sections that are too large increase installation costs without any technical need. Therefore, precise sizing of the lines is part of every professional electrical project and must comply with the regulations on low-voltage installations.

Wiring is the physical basis of the entire electrical system and connects all functional components of the restaurant into a single network. In the next chapter, we will focus on protective systems that prevent failures, protect users and ensure the correct operation of all electrical circuits in the system.

4.6 Protective systems and security

Protecting users, staff and equipment from electrical accidents is one of the key tasks of any electrical system in a restaurant. Properly designed protective systems include circuit breakers (MCBs), residual current devices (RCDs), grounding, surge protection, as well as voltage and temperature control devices. The main purpose of these systems is to prevent fires due to short circuits or overloads, protect people from electric shock and maintain the integrity of electrical circuits in the event of a network failure.

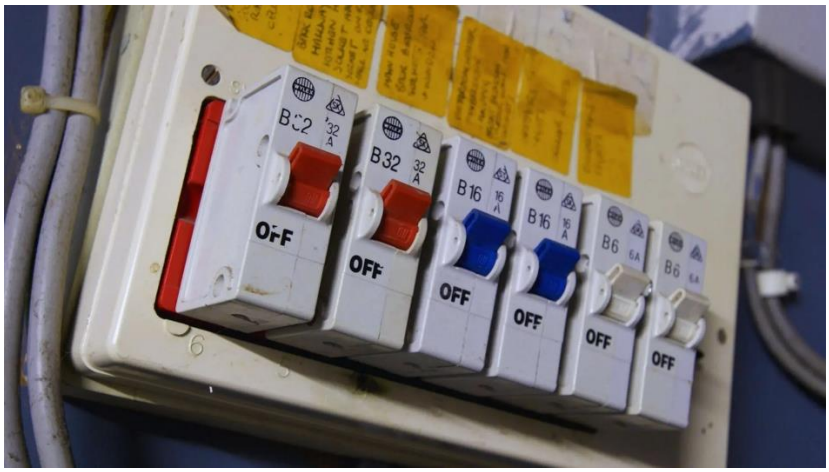


Figure 25: Circuit Breakers

Circuit breakers protect electrical circuits from excessive current and short circuits. Each circuit - be it lighting, sockets or specific devices - must have its own switch, the rated capacity of which is carefully selected based on the load and wire cross-section. Differential circuit breakers react to leakage of current into the ground, which is usually a sign of contact of the electrical installation with a person or moisture. In restaurants, the use of RCD devices with a sensitivity threshold of 30 mA is recommended for areas with the presence of people, while higher thresholds (100–300 mA) are used for fire protection.

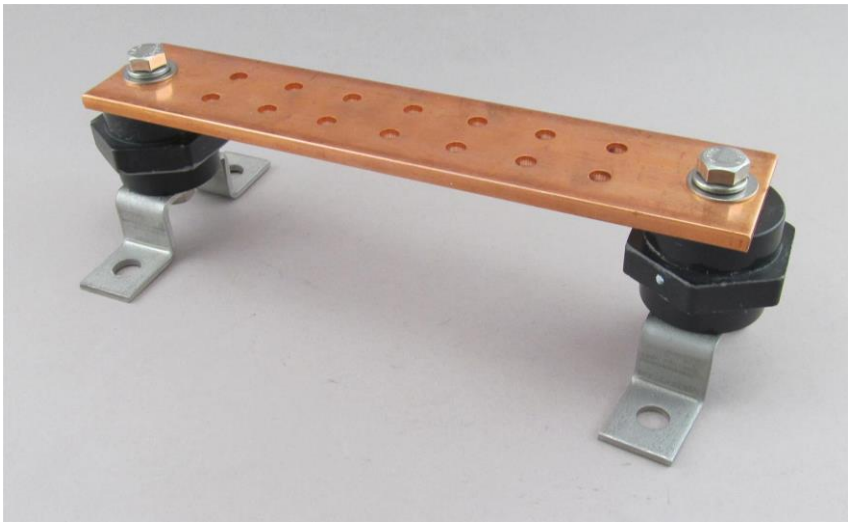


Figure 26: Grounding Bar

Grounding is a fundamental component of any installation - it allows the rapid removal of dangerous voltage differences between the metal parts of the equipment and the ground, thereby preventing

electric shock. All metal parts of devices, switchboards and installations must be connected to the grounding system, which is carried out using copper rods or strips buried in the ground and connected to the electrical installation through a protective conductor (yellow-green color). In rooms with increased humidity, such as kitchens and toilets, the grounding rule is applied with special emphasis.

Figure 27: SPDs

Surge Protection Devices (SPDs) are used to protect against lightning strikes or sudden surges in the network, which can damage sensitive electronic devices such as cash registers, lighting and control units. These devices are installed in switchboards and drain excess energy into the ground. Voltage monitoring relays can also be used, which turn off devices if the voltage drops below or rises above a safe limit.

The safety of a restaurant's electrical system depends not only on the components, but also on regular maintenance, periodic insulation measurements, connection checks and updating of electrical documentation. Every electrical system must have a valid design and be executed in accordance with national standards and regulations. The next chapter will discuss ventilation and air conditioning, as systems that depend on a reliable electrical infrastructure and further affect safety and comfort.

4.7 Ventilation and air conditioning

Ventilation and air conditioning are key systems in restaurants as they directly affect air quality, guest comfort and staff safety. These systems rely on a reliable electrical infrastructure to ensure constant air circulation and treatment in rooms for various purposes. Industrial fans, extractors and hoods are used in kitchens to remove steam, smoke, moisture and grease, while air conditioning systems are used in guest rooms to regulate temperature, humidity and air cleanliness. All of these devices require a stable power supply and often operate continuously throughout the working day.

Figure 28: Restaurant Kitchen Ventilation

Kitchen ventilation systems include powerful motors and often use three-phase current. They must be connected to electrical circuits with appropriate cable cross-sections (usually 4 mm² or 6 mm²), protective switches and automatic start-up control. Fans are often combined with temperature, smoke and humidity sensors, which automatically regulate the rotation speed to ensure effective removal of polluted air. Some systems also use time-controlled automation (timers), especially when it is necessary for ventilation to continue to operate after the end of food preparation, for additional purification of the space.

Air conditioners in guest rooms are used to maintain a comfortable temperature in the closed space and are often connected to air filtration systems. Modern air conditioners (split and multi-split systems) use electronic control units and temperature sensors, and are powered by special circuits that are separately insured in the distribution panel. Their operation must comply with energy efficiency requirements, so inverter devices that optimize electricity consumption are increasingly used. In larger restaurants, centralized HVAC systems require even more complex electrical installations, with their own control systems, automation and sensors.

The electrical components of these systems must be protected from moisture, grease and mechanical damage, which means that cables are laid in protective pipes, IP-protected connectors are used, and the equipment must be resistant to heat and corrosion. In some cases, an uninterruptible power supply (UPS) is also used for control systems, to maintain air circulation even in the event of a short-term power outage.

Ventilation and air conditioning are examples of systems whose electrical part must be carefully designed to ensure both functionality and safety. In the rest of the report, we consider the lighting system for outdoor and security purposes as a separate category within the electrical circuits of a restaurant.

4.8 Automation and System Management

Automation and management of electrical systems in restaurants are becoming increasingly widespread in order to increase efficiency, energy savings and safety levels. Although simple switches and manual control systems are often used in smaller restaurants, modern establishments are increasingly introducing smart technologies that enable centralized control of lighting, air conditioning, ventilation and other systems. Such systems are based on sensors, timers and microcontrollers that automate the operation of devices according to the needs of the establishment.

For example, lighting can be automatically adjusted to daylight or the presence of people in the space, thereby achieving significant electricity savings. Ventilation systems can operate according to preset modes or respond to humidity and temperature levels, while air conditioning can be programmed for different time periods. This automation not only reduces operating costs and consumption, but also improves guest comfort and the safety of the establishment.

Figure 29: Restaurant Control Panel

Management can be achieved via a central control panel, remote controls or even mobile applications that allow remote monitoring and control of the system. Smart systems often have integrated alarm functions that notify staff of malfunctions or irregularities in operation, which allows for timely intervention and reduces the risk of accidents.

The implementation of such systems requires careful planning of the electrical installation with special emphasis on safety, power stability and device compatibility. The installation of automation requires additional components such as relays, motion sensors, light sensors and control units, which are connected to a central system. Despite the higher initial costs, automation brings long-term benefits in terms of economy and functionality of catering systems.

The following part will present basic guidelines for the maintenance and service of electrical installations in restaurants, which ensures a long service life and safety of the system.

4.9 Maintenance and service of electrical installations

Regular maintenance and service of electrical installations in restaurants is crucial for the safety, reliability and longevity of the system. Electrical installations are exposed to daily stresses, changes in temperature, humidity and mechanical influences, especially in kitchens and technical rooms. Therefore, it is necessary to periodically check the condition of cables, switches, sockets, distribution cabinets and protective devices, in order to detect potential faults in time and prevent serious incidents.

It is recommended that the installation be inspected at least once a year, and in areas with increased risk (kitchens, boiler rooms) more often. The inspection includes measuring insulation, checking grounding, functionality of protective switches and differential devices, as well as a visual inspection of all elements. Faulty or damaged components must be replaced or repaired immediately, and all work should be carried out by qualified electricians.

It is important to keep records of all services and repairs, which allows monitoring the condition of the system and planning preventive interventions. Employees should also be provided with basic information on the correct use of electrical devices and equipment, as well as procedures in the event of an electrical fault or fire. Staff education contributes to reducing risks and increasing safety at work.

Maintaining electrical systems in restaurants is not only a technical task but also a legal obligation, in accordance with regulations and standards on occupational safety. Adhering to recommended procedures reduces the costs of unplanned breakdowns and increases business efficiency. Quality and timely care of installations contributes to creating a safe environment for employees and guests.

In conclusion, electrical systems in restaurants are complex and require careful design, implementation and maintenance. This report highlights the basic elements and rules that ensure their safety and functionality, which is of great importance for any modern hospitality facility.

4.10 Electrical Challenges And Problems Faced by Restaurants

Like any other system, this system is subject to problems that we can recognize and change or minimize only if we are well acquainted with them and see them and spot them in time.

4.10.1 Power outages

Power outages in restaurants can be a big problem and lead to much bigger problems than power outages in people's homes, because they affect earnings, depending on how long they last. Shorter outages can affect the business negatively if we cannot use ovens and other appliances needed for food preparation, or for a longer period if food starts to spoil and the like. For this reason, it is recommended that each restaurant has additional sources of energy, aggregates and the like, or that it has smaller appliances that run on some other source, such as gas.

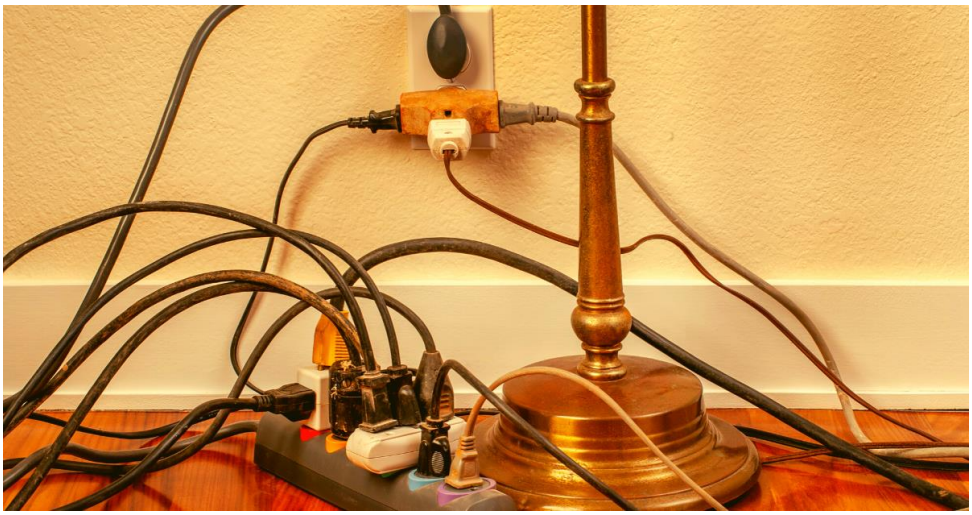


Figure 30: Overloaded Circuits

4.10.2 Overloaded Circuits

Overloaded circuits can be a major safety hazard, especially in restaurants that require large capacity for various ovens, refrigerators, fryers, and other appliances that require large amounts of energy. The only way to resolve this is to work with experts and consult with them about any changes to the kitchen.

Figure 31: Outdated Installation

4.10.3 Outdated installations

In addition to other problems that can cause great damage to property and loss of human life, it is important to mention outdated installations. When moving into any building or choosing a building for business, you should consult with experts if everything is okay with the installations, because outdated and damaged installations can be the source and cause of fires and other tragedies that can affect restaurants as well as all other buildings that use electricity.

5. Conclusion

Electrical installations form the pillar of modern restaurants, as they encompass everything from lighting and powering kitchen equipment, to air supply and cooling systems, to security features such as flame detectors and alarm systems. Without properly installed electrical installations, one cannot develop the optimum working and living environment for employees and visitors. All along throughout this project, emphasis was given to the application of the basic electrical components in a model restaurant system, which demonstrated the need for careful planning, appropriate material selection and prudent implementation.

With the design of the lighting system, operation of DC motors for air ventilation and protection through flame detectors and buzzers, issues that arise in actual environments, i.e., in power supply and surge or inductive shock protection of the devices, were beautifully highlighted. For example, the failure to directly drive a DC motor using a power bank due to the limitation of current capacity indicates how important it is to understand the nature of the power supply and the load. The fact that protective diodes are utilized in motors also confirms how important it is to protect electrical devices from destruction by inductive voltages, which is very important in any real system.

The real circumstances in Bosnia and Herzegovina, when materials and equipment are typically scarce, make the work of electrical installations even more challenging, yet also provoke imagination and inventiveness in seeking practical solutions. The project proves that basic electric components, properly connected and shielded, are able to ensure the functionality and safety of an electric system in restaurant environments even without complex control devices such as microcontrollers.

Lastly, the gained practical experience forms a suitable foundation for further development of lengthy knowledge and skill in electrical circuits and their employment in real conditions. The ability to design and employ electrical systems in diminished but functional models is crucial to being prepared to work on more complex projects in professional work. This approach is not just crucial to students, but to everyone involved with the planning and maintenance of electrical installations within homes and can contribute to safer, more efficient and more reliable operation of electrical equipment in restaurants throughout the region.

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